

LAPORAN PRAKTIKUM 6
SISTEM BASIS DATA



Disusun Oleh :

Muhammad Harits Arrozzaq (2400016087)

Kelas Praktikum/Semester :

B/III

PROGRAM STUDI SISTEM INFORMASI
FAKULTAS SAINS & TEKNOLOGI TERAPAN
UNIVERSITAS AHMAD DAHLAN
YOGYAKARTA
2025

A. Dasar Teori

Master-Slave Replication adalah mekanisme pada sistem basis data di mana satu server bertindak sebagai master dan server lainnya berperan sebagai slave. Pada metode ini, seluruh perubahan data (write operation) seperti INSERT, UPDATE, dan DELETE yang terjadi di master akan direkam dalam binary log. Server slave kemudian membaca binary log tersebut melalui proses I/O Thread, lalu mengeksekusinya kembali menggunakan SQL Thread sehingga data pada slave menjadi konsisten dan mengikuti kondisi terkini dari master.

Tujuan utama replikasi adalah meningkatkan ketersediaan (availability), menyebarkan beban kerja (load balancing), serta menyediakan backup real-time tanpa mengganggu operasi pada server utama. Dalam implementasinya, replikasi dianggap berjalan normal apabila status `Slave_IO_Running` dan `Slave_SQL_Running` bernilai *Yes* serta nilai `Seconds_Behind_Master` kecil atau mendekati nol, yang menandakan bahwa proses sinkronisasi berjalan tanpa delay signifikan.

B. Langkah – Langkah Praktikum

Master	: Yusuf Ariffandi (2411016036)
Slave	: Muhammad Harits Arrozzaq (2400016087)

1. Edit the MySQL configuration

```
1  [client]
2  #password=your_password
3  port=3306
4  socket=/tmp/mysql.sock
5
6  [mysqld]
7  server-id = 1
8  bind-address=0.0.0.0
9  log-bin =D:/PRAKTIKUM_6/log-bin.log
10 binlog_format = ROW
11 datadir="C:/laragon/data/mysql-8.4"
12 port=3306
13 socket=/tmp/mysql.sock
14 key_buffer_size=256M
15 max_allowed_packet=512M
16 table_open_cache=256
17 sort_buffer_size=1M
18 read_buffer_size=1M
19 read_rnd_buffer_size=4M
20 myisam_sort_buffer_size=64M
21 thread_cache_size=8
22
23 secure-file-priv=""
24 explicit_defaults_for_timestamp=1
25
26
27 [mysqldump]
28 quick
29 max_allowed_packet=512M
30
```

2. Create replication user

```
C:\laragon\www
mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.3 MySQL Community Server - GPL

Copyright (c) 2000, 2024, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create user '2411016036'@'%' identified by '12345';
ERROR 1396 (HY000): Operation CREATE USER failed for '2411016036'@'%'
mysql> show users;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'users' at line 1
mysql> drop user '2411016036'@'%' ;
Query OK, 0 rows affected (0.01 sec)

mysql> create user '2411016036'@'%' identified by '12345';
Query OK, 0 rows affected (0.02 sec)

mysql> grant replication slave on *.* to '2411016036'@'%' ;
Query OK, 0 rows affected (0.01 sec)

mysql> flush privileges;
Query OK, 0 rows affected (0.01 sec)
```

3. Get master coordinate

```
mysql> show binary log status;
+-----+-----+-----+-----+-----+
| File | Position | Binlog_Do_DB | Binlog_Ignore_DB | Executed_Gtid_Set |
+-----+-----+-----+-----+-----+
| PRAKTIKUM_6.000003 | 1055 | | | |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

4. Show binary log

```
mysql> show global variables like 'log_bin';
+-----+-----+-----+
| Variable_name | Value |
+-----+-----+-----+
| log_bin       | ON    |
+-----+-----+-----+
1 row in set (0.04 sec)
```

```
mysql> show global variables like '%log_bin%';
+-----+-----+-----+
| Variable_name | Value |
+-----+-----+-----+
| log_bin       | ON    |
| log_bin_basename | C:\Users\yusuf\OneDrive\Tulisan Gua\PRAKTIKUM\SBD\PRAKTIKUM_6 |
| log_bin_index  | C:\Users\yusuf\OneDrive\Tulisan Gua\PRAKTIKUM\SBD\PRAKTIKUM_6.index |
| log_bin_trust_function_creators | OFF |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> Show binary logs;\
```

Log_name	File_size	Encrypted
PRAKTIKUM_6.000003	1216	No

```
1 row in set (0.00 sec)
```

5. Lock the tables

```
mysql> flush tables with read lock;
Query OK, 0 rows affected (0.01 sec)
```

6. Show master status

```
mysql> SHOW BINARY LOG STATUS;
```

File	Position	Binlog_Do_DB	Binlog_Ignore_DB	Executed_Gtid_Set
PRAKTIKUM_6.000003	1216			

```
1 row in set (0.00 sec)
```

7. Copy the binary log file into slave pc

Sudah

8. Prepare Slave Server

```
Last login: Mon Dec 1 15:57:28 on console
(base) macbookpro@MacBookPro: ~ % which mysql
/usr/bin/mysql
(base) macbookpro@MacBookPro: ~ % ps aux | grep mysqld
macbookpro  840  0.0  0.0 35003184  2040  ??  S   7:59PM   0:29.53 /usr/local/opt/mysql/bin/mysqld --basedir=/usr/local/opt/mysql --datadir=/usr/local/var/mysql --plugin-dir=/usr/local/opt/mysql/lib/plugin --log-e
rror=MacBookPro:~:local.err --pid-file=MacBookPro:~:local.pid
macbookpro  703  0.0  0.0 34261592    0  ??  S   7:58PM   0:00.00 /bin/sh /usr/local/opt/mysql/bin/mysqld_safe --datadir=/usr/local/var/mysql
macbookpro  4592  0.0  0.0 34222144  684 s000  R   7:27PM   0:00.00 grep mysqld
(base) macbookpro@MacBookPro: ~ % sudo nano /usr/local/etc/my.cnf
Password:
(base) macbookpro@MacBookPro: ~ % brew services restart mysql
Stopping 'mysql'... (might take a while)
=> Successfully stopped 'mysql' (label: homebrew.mxcl.mysql)
=> Successfully started 'mysql' (label: homebrew.mxcl.mysql)
```

9. Set the master configuration

```
[mysql> STOP REPLICA;  
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> CHANGE REPLICATION SOURCE TO  
-> SOURCE_HOST='192.168.1.30',  
-> SOURCE_USER='2411016036',  
-> SOURCE_PASSWORD='12345',  
-> SOURCE_LOG_FILE='PRAKTIKUM_6.000003',  
-> SOURCE_LOG_POS=1216;  
Query OK, 0 rows affected, 2 warnings (0.01 sec)
```

10. Start Slave & Show Slave Status

```
[mysql> start replica;  
Query OK, 0 rows affected (0.02 sec)
```

```
[mysql> SHOW REPLICA STATUS\G  
***** 1. row *****  
Replica_IO_State: Checking source version  
Source_Host: 192.168.1.30  
Source_User: 2411016036  
Source_Port: 3306  
Connect_Retry: 60  
Source_Log_File: PRAKTIKUM_6.000003  
Read_Source_Log_Pos: 1216  
Relay_Log_File: MacBook-Air-Muhammad-relay-bin.000001  
Relay_Log_Pos: 4  
Relay_Source_Log_File: PRAKTIKUM_6.000003  
Replica_IO_Running: Yes  
Replica_SQL_Running: Yes  
Replicate_Do_DB:  
Replicate_Ignore_DB:  
Replicate_Do_Table:  
Replicate_Ignore_Table:  
Replicate_Wild_Do_Table:  
Replicate_Wild_Ignore_Table:  
Last_Errno: 0  
Last_Error:  
Skip_Counter: 0  
Exec_Source_Log_Pos: 1216  
Relay_Log_Space: 158  
Until_Condition: None  
Until_Log_File:  
Until_Log_Pos: 0  
Source_SSL_Allowed: Yes  
Source_SSL_CA_File:  
Source_SSL_CA_Path:  
Source_SSL_Cert:  
Source_SSL_Cipher:  
Source_SSL_Key:  
Seconds_Behind_Source: 0  
Source_SSL_Verify_Server_Cert: No  
Last_IO_Errno: 0  
Last_IO_Error:  
Last_SQL_Errno: 0  
Last_SQL_Error:  
Replicate_Ignore_Server_Ids:  
Source_Server_Id: 1  
Source_UUID:  
Source_Info_File: mysql.slave_master_info  
SQL_Delay: 0  
SQL_Remaining_Delay: NULL  
Replica_SQL_Running_State: Replica has read all relay log; waiting for more updates  
Source_Retry_Count: 10  
Source_Bind:  
Last_IO_Error_Timestamp:  
Last_SQL_Error_Timestamp:  
Source_SSL_Crl:  
Source_SSL_Crlpath:  
Retrieved_Gtid_Set:  
Executed_Gtid_Set:  
Auto_Position: 0  
Replicate_Rewrite_DB:  
Channel_Name:  
Source_TLS_Version:  
Source_public_key_path:  
Get_Source_public_key: 1  
Network_Namespace:  
1 row in set (0.006 sec)
```

11. Make sure replication work well with write DDL or DML query

```
[mysql> unlock tables;  
Query OK, 0 rows affected (0.00 sec)
```

12. Testing for replication work

```
mysql> create database yunus;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> use yunus;  
Database changed
```

```
mysql> create table mahasiswa (  
-> nama varchar (100),  
-> nim varchar (100),  
-> kelamin varchar (100)  
-> );  
Query OK, 0 rows affected (0.18 sec)
```

```
mysql> INSERT INTO mahasiswa (nama, nim, kelamin) VALUES ('Muhammad Harits Arrozzaq', '2400016087', 'L'), ('Muhammad Fauzi Al Bukhari', '2400016073', 'L'), ('Yusuf Ariffandi', '2411016036', 'L');  
Query OK, 3 rows affected (0.01 sec)
```

```
mysql> select * from mahasiswa;  
+-----+-----+-----+  
| nama          | nim          | kelamin |  
+-----+-----+-----+  
| Muhammad Harits Arrozzaq | 2400016087 | L       |  
| Muhammad Fauzi Al Bukhari | 2400016073 | L       |  
| Yusuf Ariffandi          | 2411016036 | L       |  
+-----+-----+-----+  
3 rows in set (0.00 sec)
```

13. Make data change on the master

```
C:\laragon\www
λ mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.4.3 MySQL Community Server - GPL

Copyright (c) 2000, 2024, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use yunus;
Database changed
mysql> INSERT INTO mahasiswa (nama, nim, kelamin) VALUES
-> ('Fernando', 240001232, 'L')
-> ;
Query OK, 1 row affected (0.07 sec)

mysql> UPDATE mahasiswa SET nim = '2412345678' WHERE nama = 'Fernando';
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> DELETE FROM mahasiswa WHERE nama = 'Fernando';
Query OK, 1 row affected (0.04 sec)

mysql> exit
Bye
```

14. Verify the data change on the master

```
C:\laragon\www
λ mysql -u 2411061036 -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.4.3 MySQL Community Server - GPL

Copyright (c) 2000, 2024, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use yunus
Database changed
mysql> select*from mahasiswa;
+-----+-----+-----+
| nama                | nim      | kelamin |
+-----+-----+-----+
| Muhammad Harits Arrozzaq | 2400016087 | L       |
| Muhammad Fauzi Al Bukhari | 2400016073 | L       |
| Yusuf Ariffandi      | 2411016036 | L       |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

15. Check the Slave Status

```
mysql> SHOW REPLICA STATUS\G
***** 1. row *****
Replica_IO_State: Checking source version
Source_Host: 192.168.1.30
Source_User: 2411816036
Source_Port: 3306
Connect_Retry: 60
Source_Log_File: PRAKTIKUM_6.000003
Read_Source_Log_Pos: 1216
Relay_Log_File: MacBook-Air-Muhammad-relay-bin.000001
Relay_Log_Pos: 4
Relay_Source_Log_File: PRAKTIKUM_6.000003
Replica_IO_Running: Yes
Replica_SQL_Running: Yes
Replicate_Do_DB:
Replicate_Ignore_DB:
Replicate_Do_Table:
Replicate_Ignore_Table:
Replicate_Wild_Do_Table:
Replicate_Wild_Ignore_Table:
Last_Errno: 0
Last_Error:
Skip_Counter: 0
Exec_Source_Log_Pos: 1216
Relay_Log_Space: 158
Until_Condition: None
Until_Log_File:
Until_Log_Pos: 0
Source_SSL_Allowed: Yes
Source_SSL_CA_File:
Source_SSL_CA_Path:
Source_SSL_Cert:
Source_SSL_Cipher:
Source_SSL_Key:
Seconds_Behind_Source: 0
Source_SSL_Verify_Server_Cert: No
Last_IO_Errno: 0
Last_IO_Error:
Last_SQL_Errno: 0
Last_SQL_Error:
Replicate_Ignore_Server_Ids:
Source_Server_Id: 1
Source_UUID:
Source_Info_File: mysql.slave_master_info
SQL_Delay: 0
SQL_Remaining_Delay: NULL
Replica_SQL_Running_State: Replica has read all relay log; waiting for more updates
Source_Retry_Count: 10
Source_Bind:
Last_IO_Error_Timestamp:
Last_SQL_Error_Timestamp:
Source_SSL_Crl:
Source_SSL_Crlpath:
Retrieved_Gtid_Set:
Executed_Gtid_Set:
Auto_Position: 0
Replicate_Rewrite_DB:
Channel_Name:
Source_TLS_Version:
Source_public_key_path:
Get_Source_public_key: 1
Network_Namespace:
1 row in set (0.006 sec)
```

C. Kesimpulan

Replikasi MySQL memungkinkan data pada server master tersalin secara otomatis ke server slave. Untuk memastikan proses replikasi berjalan dengan benar, lakukan perubahan data pada master (misalnya INSERT, UPDATE, atau DELETE), lalu periksa status slave menggunakan perintah **SHOW SLAVE STATUS\G**. Replikasi dinyatakan berhasil apabila **Slave_IO_Running** dan **Slave_SQL_Running** bernilai *Yes*, serta **Seconds_Behind_Master** mendekati nol. Langkah ini penting sebagai bagian dari validasi praktikum maupun demo teknis agar peserta memahami alur sinkronisasi antar-server secara real-time.